

TIQCollect -- Machine Learning Architecture

Empirical Bayes Shrinkage & Global Bipartite Linear Programming for Case Allocation

1. Executive Summary & The Problem

Debt collection operations face a fundamental trade-off: allocating delinquent loans to field agents in a way that maximizes total recovered cash while minimizing travel expenses, respecting agent capacity limits, and avoiding customer friction. Traditional collection systems rely on **greedy heuristic allocation** (e.g., matching the closest agent to a case one-by-one). This naive approach leads to severe operational bottlenecks, unfair agent workloads, and millions in lost recoveries.

Flaw in Legacy Systems	Real-World Consequence	TIQCollect Solution
Greedy 1-by-1 Matching	First-come assignments grab local easy cases, leaving distant high-value cases stranded.	Global Bipartite Matcher: Optimizes all cases and agents simultaneously.
Direct Case+Agent ML Bias	Models learn past assignment bias; lucky agents get easy cases, creating a feedback loop.	Empirical Bayes Shrinkage: Multi-level borrowing of segment priors.
Ignoring Rupee Values	Agents travel 25 km to collect Rs. 2,000 while an adjacent Rs. 1,50,000 account is neglected.	Expected INR Formulation: Weights allocation by true expected recovery amount.
Single Rigid Objective	Managers cannot adjust operational strategy for fuel savings vs. month-end cash targets.	Multi-Objective Engine: Max Recovery, Balanced, or Min Distance at a click.

2. System Architecture Overview

TIQCollect separates case allocation into a **Two-Stage Decoupled AI Pipeline** that blends statistical machine learning with deterministic linear programming constraints:

Two-Stage Mathematical Framework
Stage 1: Statistical Learning (Empirical Bayes & Shadow Evaluator) - Historical repayments are partitioned into (<i>Loan Type x DPD Bucket</i>) risk segments. - Agent historical performance is estimated with Beta-Binomial Bayesian shrinkage to eliminate small-sample noise. - Calculates the True Recovery Probability $P(\text{Recovery} \mid \text{Agent}, \text{Case})$ for all possible pairings.
Stage 2: Global Bipartite Linear Programming (Scipy linear_sum_assignment) - Constructs an $N \times M$ global utility matrix combining Expected Rupee Value, Proximity, Skill, and Workload. - Solves the optimal global assignment globally in < 25 ms under manager-selected objectives. - Validates travel feasibility with OR-Tools route radius gates; defers route-infeasible cases cleanly.

3. The Core ML Model: Empirical Bayes Shrinkage

A critical question in collection AI is: *Why not train a standard XGBoost or Deep Neural Network predicting recovery on (Agent ID + Case ID)?*

Why Standard Black-Box ML Fails Here

Why Direct Case+Agent ML Models Fail in Production:

1. **Past Assignment Confounding (Selection Bias):** In historical data, an agent only worked the cases assigned to them. If Agent A was historically assigned easy 15-day DPD cases and Agent B was assigned hostile 90-day DPD cases, a standard ML classifier learns that 'Agent A has a 90% win rate and Agent B has a 10% win rate'. It falsely concludes Agent A should get every case.
2. **The Small Sample Distortion Trap:** If Agent C only visited 2 cases in their career and won both, their raw win rate is 100%. A naive ML model rates Agent C higher than a veteran agent who has a proven 75% win rate over 500 cases.
3. **Target Leakage & Promise Confusion:** Combining promises (PTP) and cash payments as a single target leads models to reward agents who record fake promises that never clear.

How the Empirical Bayes Beta-Binomial Model Works

To solve these fundamental statistical hazards, TIQCollect implements **Beta-Binomial Empirical Bayes Shrinkage** (backend/app/ml/empirical_bayes.py). This technique borrows statistical strength across the entire agency portfolio:

Step 1: Portfolio Segment Priors

Every historic case is assigned to a segment $s = (\text{Loan Type}, \text{DPD Bucket})$ (e.g., *Personal Loan x 61-90 DPD*). The system computes the global agency prior mean win rate μ_s across all agents.

Step 2: Bayesian Shrinkage Formula

For an individual agent j with $n_{j,s}$ historical attempts in segment s and observed win rate w_{observed} , their estimated true win rate $w_{\text{estimated}}$ is smoothed towards the segment prior:

Empirical Bayes Formulation

$$w_{\text{estimated}}(j, s) = [n / (n + M)] * (w_{\text{observed}}) + [M / (n + M)] * (\mu_{\text{segment}})$$

Where $M = 10$ is the prior shrinkage strength parameter.

- **Small Sample ($n = 2$):** $w_{\text{estimated}} = 0.17 * (\text{Observed}) + 0.83 * (\text{Segment Prior})$ -- *Shrinks heavily to segment baseline.*
- **Large Sample ($n = 40$):** $w_{\text{estimated}} = 0.80 * (\text{Observed}) + 0.20 * (\text{Segment Prior})$ -- *Relies on proven agent ability.*

Step 3: Multiplier Safety Clamping

The agent segment multiplier $\alpha(j, s) = w_{\text{estimated}} / \mu_{\text{segment}}$ is strictly bounded to the range **[0.75, 1.25]**. This guarantees that no agent is penalized or boosted by more than +/- 25% based on past history, preventing runaway star-performer bias.

4. Global Linear Programming Optimization

Once individual case-agent probabilities are estimated, TIQCollect models daily portfolio allocation as a **Bipartite Maximum-Weight Matching Problem** solved with the Hungarian / Jonker-Volgenant algorithm (scipy.optimize.linear_sum_assignment).

Expected Rupee Utility Function

For every candidate case *i* and agent capacity slot *j*, the engine calculates a multi-dimensional utility score:

Global Utility Matrix Formulation

$Utility(i, j) = w_INR * Score_INR(i, j) + w_dist * Score_prox(i, j) + w_skill * Score_skill(j) + Continuity + Language$

- **Expected Rupee Score (Score_INR):** Target Balance x P(Recovery | Agent, Segment). Scales with true recovery value.
- **Proximity Score (Score_prox):** $\exp(-Distance_km / 5.5)$. Exponential decay penalizing long commutes.
- **Skills & Tier (Score_skill):** Agent Tier weight (Tier 1 = 1.0, Tier 2 = 0.8) + Product Specialization match.
- **Capacity Decay:** Slightly prioritizes early morning slots over late fatigue slots.

Manager-Selectable Allocation Objectives

Managers can instantly switch the agency allocation strategy in the Command Center. The LP solver recalculates the entire agency portfolio in under 30 milliseconds:

Objective	Weight Formulation	Operational Impact on Active Portfolio
Max Recovery	70% INR Recovery 10% Proximity 15% Skills & Tier	Rs. 138.8 Lakhs Expected Recovery Maximizes collections on high-value delinquent accounts. Top choice for month-end closes.
Balanced	45% INR Recovery 40% Proximity 10% Skills & Tier	Rs. 136.0 Lakhs Expected Recovery Achieves 98% of peak recovery while cutting travel distance by 27% (667 km total).
Min Distance	80% Proximity 5% INR Recovery 10% Workload Balance	467.2 km Total Travel (49% Fuel Savings) Clusters cases tightly around agent homes. Ideal during bad weather or high fuel costs.

Stage 2: Travel Feasibility & Boundary Gating

After the global solver outputs candidate matches, Stage 2 enforces physical constraints: any case requiring an excessive marginal detour (> 22 km outside the agent's territorial cluster) is rejected and tagged as **DEFERRED_ROUTE_INFEASIBLE**. This prevents agents from being assigned unserviceable isolated cases.

5. Shadow ML Evaluator & Doorstep Smart Ranking

The Shadow ML Classifier (shadow_evaluator.py)

In addition to Empirical Bayes allocation, a background **Supervised Shadow Model** evaluates point-in-time loan outcome vectors (DPD progression, overdue ratio, visit history, prior payment frequency). During every planning run, this classifier computes recovery lift deciles and saves diagnostic telemetry to AllocationRun.summary_metadata['shadow_evaluation'] for auditability.

Real-Time Doorstep Case Ranking (Smart Order)

When an agent is in the field, clicking '**Smart Order**' in the mobile app switches from standard geographic route sequencing to real-time urgency prioritization (CaseService.ranked_cases):

Live Field Signal	Score Bonus	Badge	Doorstep Impact
Promise to Pay (PTP) Due Today	+60 pts	PTP TODAY	Jumps to #1 rank. Agent visits before customer changes mind.
Verbal Payment Date Today	+50 pts	PAYMENT TODAY	Confirmed on pre-visit call; customer has funds ready.
Time-of-Day Window Match	+35 pts	BEST TIME NOW	Matches customer request (e.g. 'visit after 2 PM').
Customer Visited Today	Auto-Deprioritize	COMPLETED	Automatically moved to bottom of list with checkmark.

6. Architectural Summary & Production Guarantees

Production Quality & Reliability

Key Technical Advantages of TIQCollect's ML Stack:

- Zero Selection Bias:** Empirical Bayes shrinkage prevents runaway feedback loops and handles low-visit agents safely.
- Global Portfolio Optimization:** Bipartite linear programming guarantees the agency-wide optimal allocation in < 25 ms.
- Managerial Agility:** Strategic objective switching (INR Recovery vs. Fuel Distance) without code changes or downtime.
- Auditability & Safety:** Hard compliance gates (DNC, Hostility, Female-agent constraints) are strictly non-negotiable.
- Offline Resiliency:** All beats, sequences, and case intelligence cache locally for offline doorstep execution.

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